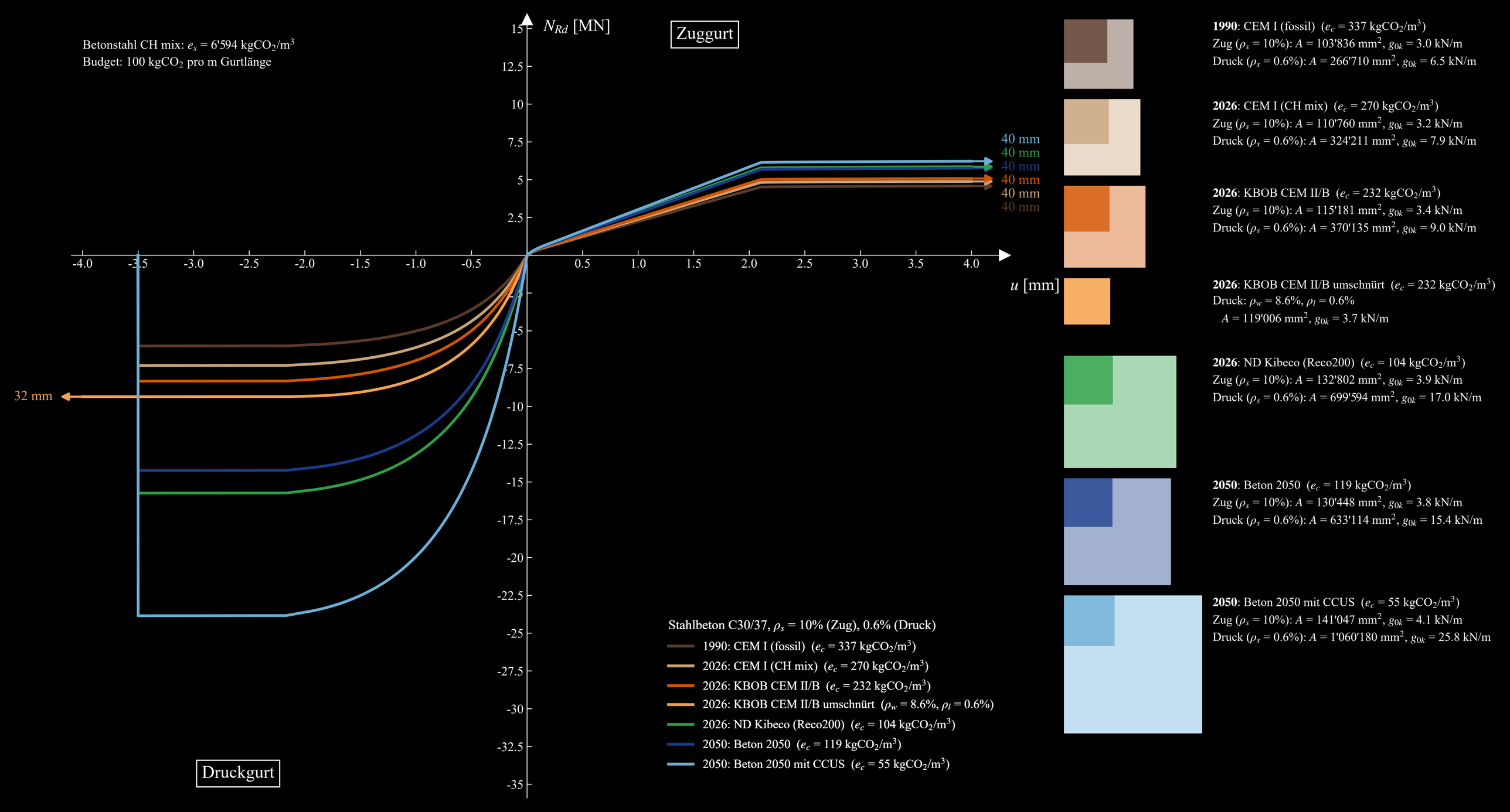

Betonstahl CH mix: e_s = 6'594 kgCO$_2$/m^3
Budget: 100 kgCO$_2$ pro m Gurtlänge
N_{Rd} [MN]
Zuggurt
Druckgurt
u [mm]
40 mm
40 mm
40 mm
40 mm
40 mm
40 mm
32 mm
15
12.5
10
7.5
5
2.5
-5
-7.5
-10
-12.5
-15
-17.5
-20
-22.5
-25
-27.5
-30
-32.5
-35
-4.0
-3.5
-3.0
-2.5
-2.0
-1.5
-1.0
-0.5
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Stahlbeton C30/37, ρ_s = 10% (Zug), 0.6% (Druck)
1990: CEM I (fossil) (e_c = 337 kgCO$_2$/m^3)
2026: CEM I (CH mix) (e_c = 270 kgCO$_2$/m^3)
2026: KBOB CEM II/B (e_c = 232 kgCO$_2$/m^3)
2026: KBOB CEM II/B umschnürt (ρ_w = 8.6%, ρ_l = 0.6%)
2026: ND Kibeco (Reco200) (e_c = 104 kgCO$_2$/m^3)
2050: Beton 2050 (e_c = 119 kgCO$_2$/m^3)
2050: Beton 2050 mit CCUS (e_c = 55 kgCO$_2$/m^3)
1990: CEM I (fossil) (e_c = 337 kgCO$_2$/m^3)
Zug (ρ_s = 10%): A = 103'836 mm^2, g_{0k} = 3.0 kN/m
Druck (ρ_s = 0.6%): A = 266'710 mm^2, g_{0k} = 6.5 kN/m
2026: CEM I (CH mix) (e_c = 270 kgCO$_2$/m^3)
Zug (ρ_s = 10%): A = 110'760 mm^2, g_{0k} = 3.2 kN/m
Druck (ρ_s = 0.6%): A = 324'211 mm^2, g_{0k} = 7.9 kN/m
2026: KBOB CEM II/B (e_c = 232 kgCO$_2$/m^3)
Zug (ρ_s = 10%): A = 115'181 mm^2, g_{0k} = 3.4 kN/m
Druck (ρ_s = 0.6%): A = 370'135 mm^2, g_{0k} = 9.0 kN/m
2026: KBOB CEM II/B umschnürt (e_c = 232 kgCO$_2$/m^3)
Druck: ρ_w = 8.6%, ρ_l = 0.6%
A = 119'006 mm^2, g_{0k} = 3.7 kN/m
2026: ND Kibeco (Reco200) (e_c = 104 kgCO$_2$/m^3)
Zug (ρ_s = 10%): A = 132'802 mm^2, g_{0k} = 3.9 kN/m
Druck (ρ_s = 0.6%): A = 699'594 mm^2, g_{0k} = 17.0 kN/m
2050: Beton 2050 (e_c = 119 kgCO$_2$/m^3)
Zug (ρ_s = 10%): A = 130'448 mm^2, g_{0k} = 3.8 kN/m
Druck (ρ_s = 0.6%): A = 633'114 mm^2, g_{0k} = 15.4 kN/m
2050: Beton 2050 mit CCUS (e_c = 55 kgCO$_2$/m^3)
Zug (ρ_s = 10%): A = 141'047 mm^2, g_{0k} = 4.1 kN/m
Druck (ρ_s = 0.6%): A = 1'060'180 mm^2, g_{0k} = 25.8 kN/m